

量子场论

zhx-23 秋-期末

1. (10 分) 在四维时空化简

$$\gamma^\mu \gamma_\mu, \quad \gamma^\mu \not{k} \gamma_\mu, \quad \gamma^\mu \not{k} \not{p} \gamma_\mu.$$

2. (20 分) 给定自由费米子拉氏量

$$\mathcal{L} = i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi.$$

- (1) 求 ψ 所满足的运动方程;
- (2) 证明 ψ 同时满足 Klein-Gordon 方程。

3. (25 分) 给定自由复标量场拉氏量

$$\mathcal{L} = \partial^\mu\Phi\partial_\mu\Phi^* - m^2\Phi\Phi^*.$$

- (1) 写下 Φ 和 Φ^* 的产生湮灭算符展开;
- (2) 写下 Φ 和 Φ^* 应满足的等时对易关系, 并写下产生湮灭算符的所有对易关系;
- (3) 写下该拉氏量所具有的整体内部对称性及相应守恒流;
- (4) 求复标量场的运动方程, 并用其验证上一小题的流守恒;
- (5) 将守恒流时间分量的全空间积分用产生湮灭算符表示, 并说明其意义。

4. (30 分) 通过极小耦合引入复标量场与电磁场的相互作用:

$$\mathcal{L} = D^\mu\Phi(D_\mu\Phi)^* - m^2\Phi\Phi^* - \frac{1}{4}F_{\mu\nu}F^{\mu\nu},$$

其中

$$D_\mu = \partial_\mu + ieA_\mu, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

- (1) 写下定域规范对称性及对应场变换;
- (2) 写下动量空间费曼规则, 包括传播子、相互作用顶点和外线极化因子;
- (3) 树图水平计算 $\Phi(p_1) + \Phi^*(p_2) \rightarrow \gamma(k_1) + \gamma(k_2)$ 的散射振幅 $\mathcal{M}$;
- (4) 将振幅写成 $\mathcal{M} = M_\mu \epsilon^{\mu,*}(k_1)$, 具体计算 $M_\mu k_1^\mu$;
- (5) 单圈水平画出 $\gamma(p_1) + \gamma(p_2) \rightarrow \gamma(k_1) + \gamma(k_2)$ 的所有费曼图并写下振幅, 无需求积分;
- (6) 计算上一小问振幅的发散项系数。

5. (15 分) 考虑定义在五维时空的 $U(1)$ 规范理论

$$S = \int d^5x \left[-\frac{1}{4}F_{MN}F^{MN} \right],$$

其中

$$x^M = (x^\mu, x^5), \quad M = 0, 1, 2, 3, 5, \quad \mu = 0, 1, 2, 3,$$

$$F_{MN} = \partial_M A_N - \partial_N A_M, \quad A_M \rightarrow A_M + \partial_M \alpha.$$

将第五维紧致化到半径为 R 的圆:

$$x^5 = R\phi, \quad \phi \in [0, 2\pi],$$

并展开

$$A_M(x^\mu, \phi) = A_M^{(0)}(x^\mu) + \sum_{n=1}^{\infty} \left(A_M^{(n)}(x^\mu) e^{in\phi} + \text{h.c.} \right).$$

- (1) 证明对非零 n 可通过规范变换令 $A_5^{(n)} = 0$, 求相应规范函数, 并说明为何 $A_5^{(0)}$ 无法如此去掉;
- (2) 当 R 远小于可观测尺度时, 积掉第五维, 写出四维理论拉氏量并给出粒子谱 (质量与自旋)。

原卷附：可能有用的公式

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}.$$

$$\frac{1}{A_1 A_2 \cdots A_n} = \int_0^1 dx_1 \cdots dx_n \delta\left(\sum_i x_i - 1\right) \frac{(n-1)!}{(x_1 A_1 + \cdots + x_n A_n)^n}.$$

$$\int \frac{d^d \ell}{(2\pi)^d} \frac{1}{(\ell^2 - \Delta)^n} = \frac{(-1)^n i}{(4\pi)^{d/2}} \frac{\Gamma(n - d/2)}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n-d/2}.$$

$$\int \frac{d^d \ell}{(2\pi)^d} \frac{\ell^2}{(\ell^2 - \Delta)^n} = \frac{(-1)^{n-1} i}{(4\pi)^{d/2}} \frac{d}{2} \frac{\Gamma(n - d/2 - 1)}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n-d/2-1}.$$

$$\int \frac{d^d \ell}{(2\pi)^d} \frac{\ell^\mu \ell^\nu}{(\ell^2 - \Delta)^n} = \frac{(-1)^{n-1} i}{(4\pi)^{d/2}} \frac{g^{\mu\nu}}{2} \frac{\Gamma(n - d/2 - 1)}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n-d/2-1}.$$

$$\int \frac{d^d \ell}{(2\pi)^d} \frac{(\ell^2)^2}{(\ell^2 - \Delta)^n} = \frac{(-1)^n i}{(4\pi)^{d/2}} \frac{d(d+2)}{4} \frac{\Gamma(n - d/2 - 2)}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n-d/2-2}.$$

$$\begin{aligned} & \int \frac{d^d \ell}{(2\pi)^d} \frac{\ell^\mu \ell^\nu \ell^\rho \ell^\sigma}{(\ell^2 - \Delta)^n} \\ &= \frac{(-1)^n i}{(4\pi)^{d/2}} \frac{\Gamma(n - d/2 - 2)}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n-d/2-2} \frac{1}{4} (g^{\mu\nu} g^{\rho\sigma} + g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho}). \end{aligned}$$

$$\Gamma(\epsilon) = \frac{1}{\epsilon} - \gamma + \mathcal{O}(\epsilon).$$

量子场论

zhx-25 秋-期末

1. (40 分) 四维单个复标量场的可重整拉格朗日量密度为

$$\mathcal{L} = \partial_\mu \Phi^\dagger \partial^\mu \Phi - m^2 \Phi^\dagger \Phi - \frac{\lambda}{4} (\Phi^\dagger \Phi)^2.$$

- (1) 写下所有费曼规则;
- (2) 写下所有非零 $2 \rightarrow 2$ 过程, 画出相应费曼图, 并计算相应树图散射振幅;
- (3) 画出上述过程的所有单圈单粒子不可约费曼图, 并在维数正规化下算出相应发散项。

2. (40 分) 标量 QED。

- (1) 将第一题理论耦合到 Maxwell 场, 写出标量 QED 的完整拉格朗日量密度;
- (2) 在费曼规范下写出该理论所有费曼规则;
- (3) 计算树图过程

$$\Phi(p_1) + \Phi^*(p_2) \rightarrow \gamma(k_1) + \gamma(k_2),$$

写成 $\mathcal{M} = \mathcal{M}_{\mu\nu} \epsilon^{\mu,*}(k_1) \epsilon^{\nu,*}(k_2)$, 并具体检验 Ward 恒等式。

3. (20 分) 考虑模型

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \sigma)^2 + i \sum_{j=1}^{N_f} \bar{\psi}_j \not{\partial} \psi_j - \sum_{j=1}^{N_f} g_1 \sigma \bar{\psi}_j \psi_j - \frac{1}{4!} g_2 \sigma^4,$$

其中 σ 为实标量场, $\psi_j (j = 1, \dots, N_f)$ 为 N_f 个狄拉克场。采用维数正规化 $D = 4 - 2\epsilon$ 和 MS 方案。

- (1) 定义重整化场 $\sigma = \sqrt{Z_\sigma} \sigma_r$ 、 $\psi_j = \sqrt{Z_\psi} \psi_{r,j}$ 和重整化耦合

$$g_1 = \mu^\epsilon Z_1 g_{r,1}, \quad g_2 = \mu^{2\epsilon} Z_2 g_{r,2}.$$

写出重整化微扰论下的拉格朗日密度及所有费曼规则 (含抵消项);

- (2) 计算单圈场重整化常数;
- (3) 计算 $g_{r,1}^2$ 和 $g_{r,2}$ 的单圈 beta 函数;
- (4) 利用 $4 - 2\epsilon$ 展开, 取 $N_f = 1/4$, 尽可能找出 $g_{r,2} > 0$ 的 beta 函数零点并判断红外稳定性, 在 $g_{r,1}^2 - g_{r,2}$ 平面画出 RG 流。

原卷附：2025 PKU QFT-1 Cheat Sheet

Free real scalar field

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2.$$

$$\phi(x) = \int \frac{d^3 p}{(2\pi)^3 2E_p} \left(a_p e^{-ip \cdot x} + a_p^\dagger e^{ip \cdot x} \right).$$

$$[a_p, a_{p'}^\dagger] = (2\pi)^3 2E_p \delta^{(3)}(\mathbf{p} - \mathbf{p}').$$

$$a_p |0\rangle = 0, \quad |\mathbf{p}\rangle = a_p^\dagger |0\rangle.$$

若无穷小变换 $\phi \rightarrow \phi + \epsilon \delta \phi$ 使 $\mathcal{L} \rightarrow \mathcal{L} + \epsilon \partial_\mu F^\mu$, 则

$$j^\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi - F^\mu.$$

平移对称给出

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial^\nu \phi - g^{\mu\nu} \mathcal{L}, \quad P^\mu = \int d^3 x T^{0\mu} = (H, \mathbf{P}).$$

$$H = \int \frac{d^3 p}{(2\pi)^3 2E_p} E_p a_p^\dagger a_p, \quad \mathbf{P} = \int \frac{d^3 p}{(2\pi)^3 2E_p} \mathbf{p} a_p^\dagger a_p.$$

$$[P^\mu, \phi(x)] = -i \partial^\mu \phi(x), \quad \phi(x) = e^{iP \cdot x} \phi(0) e^{-iP \cdot x}.$$

Lorentz 流与荷:

$$\mathcal{J}^{\mu, \rho\sigma} = x^\rho T^{\mu\sigma} - x^\sigma T^{\mu\rho}, \quad J^{\rho\sigma} = \int d^3 x \mathcal{J}^{0, \rho\sigma}.$$

$$\langle 0 | T \phi(x) \phi(y) | 0 \rangle = \int \frac{d^4 p}{(2\pi)^4} \frac{i e^{-ip \cdot (x-y)}}{p^2 - m^2 + i\epsilon}.$$

Wick 定理示例：四场时间序乘积等于正常序项加所有单收缩与双收缩项。

Free Dirac field

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} I.$$

Weyl 基:

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}, \quad \sigma^\mu = (I, \boldsymbol{\sigma}), \quad \bar{\sigma}^\mu = (I, -\boldsymbol{\sigma}).$$

$$\mathcal{L} = \bar{\psi} (i \not{\partial} - m) \psi, \quad \bar{\psi} = \psi^\dagger \gamma^0.$$

$$(i \not{\partial} - m) \psi = 0, \quad \bar{\psi} (i \overleftarrow{\not{\partial}} + m) = 0.$$

$$\psi(x) = \int \frac{d^3 p}{(2\pi)^3 2E_p} \sum_s \left(a_p^s u^s(p) e^{-ip \cdot x} + b_p^{s\dagger} v^s(p) e^{ip \cdot x} \right).$$

$$\{a_p^r, a_q^{s\dagger}\} = \{b_p^r, b_q^{s\dagger}\} = (2\pi)^3 \delta^{rs} 2E_p \delta^{(3)}(\mathbf{p} - \mathbf{q}).$$

$$a_p^s |0\rangle = b_p^s |0\rangle = 0, \quad |p, a, s\rangle = a_p^{s\dagger} |0\rangle, \quad |p, b, s\rangle = b_p^{s\dagger} |0\rangle.$$

$$\sum_s u^s(p) \bar{u}^s(p) = \not{p} + m, \quad \sum_s v^s(p) \bar{v}^s(p) = \not{p} - m.$$

$$\bar{u}^s(p) u^r(p) = 2m \delta^{sr}, \quad \bar{v}^s(p) v^r(p) = -2m \delta^{sr}.$$

$$u^{s\dagger}(p) u^r(p) = v^{s\dagger}(p) v^r(p) = 2E_p \delta^{sr}.$$

$$T^{\mu\nu} = i \bar{\psi} \gamma^\mu \partial^\nu \psi - g^{\mu\nu} \mathcal{L}.$$

$$H = \int \frac{d^3 p}{(2\pi)^3 2E_p} E_p \sum_s (a_p^{s\dagger} a_p^s + b_p^{s\dagger} b_p^s).$$

$$J^{\mu, \alpha\beta} = x^\alpha T^{\mu\beta} - x^\beta T^{\mu\alpha} + \bar{\psi} \gamma^\mu \Sigma^{\alpha\beta} \psi, \quad \Sigma^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu].$$

$$j^\mu = \bar{\psi} \gamma^\mu \psi, \quad Q = \int d^3 x j^0,$$

$$Q = \int \frac{d^3 p}{(2\pi)^3 2E_p} \sum_s (a_p^{s\dagger} a_p^s - b_p^{s\dagger} b_p^s).$$

$$\partial_\mu T^{\mu\nu} = 0, \quad \partial_\mu J^{\mu, \alpha\beta} = 0, \quad \partial_\mu j^\mu = 0.$$

$$S_F(x-y) = \int \frac{d^4 p}{(2\pi)^4} \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon} e^{-ip \cdot (x-y)}.$$

Free $U(1)$ gauge field, Feynman gauge

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}(\partial_\mu A^\mu)^2, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

$$\partial_\mu F^{\mu\nu} + \partial^\nu (\partial \cdot A) = 0.$$

$$A^\mu(x) = \int \frac{d^3p}{(2\pi)^3 2E_p} \sum_{\lambda=0}^3 \epsilon_\lambda^\mu(p) \left(a_{\mathbf{p}}^\lambda e^{-ip \cdot x} + a_{\mathbf{p}}^{\lambda\dagger} e^{ip \cdot x} \right).$$

Discrete symmetry

Parity $P: (t, \mathbf{x}) \mapsto (t, -\mathbf{x})$:

$$\phi(t, \mathbf{x}) \rightarrow \phi(t, -\mathbf{x}) \quad (\text{scalar}),$$

$$\phi(t, \mathbf{x}) \rightarrow -\phi(t, -\mathbf{x}) \quad (\text{pseudoscalar}),$$

$$\psi(x) \rightarrow \eta_P \gamma^0 \psi(Px), \quad A^\mu \rightarrow (A^0, -\mathbf{A})(Px).$$

Time reversal $T: (t, \mathbf{x}) \mapsto (-t, \mathbf{x})$:

$$\phi(t, \mathbf{x}) \rightarrow \phi(-t, \mathbf{x}), \quad A^\mu \rightarrow (A^0, -\mathbf{A})(Tx),$$

并对自旋场使用相应反么正变换。

Charge conjugation:

$$\phi(x) \rightarrow \phi^\dagger(x), \quad \psi(x) \rightarrow -i\gamma^2 \psi^*(x), \quad A^\mu \rightarrow -A^\mu.$$

Interacting real scalar field

$$\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m_0^2\phi^2 - \frac{\lambda_0}{4!}\phi^4.$$

$$\phi_0 = \sqrt{Z_\phi}\phi, \quad m_0^2 = Z_m m^2, \quad \lambda_0 = Z_\lambda \lambda \mu^{2\epsilon}.$$

$$\begin{aligned} \mathcal{L} = & \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda\mu^{2\epsilon}}{4!}\phi^4 \\ & + \frac{1}{2}\delta_Z(\partial\phi)^2 - \frac{1}{2}\delta_m m^2\phi^2 - \delta_\lambda \frac{\lambda\mu^{2\epsilon}}{4!}\phi^4. \end{aligned}$$

反项规则: 外线标量 $1; \phi^4$ 顶点 $-i\lambda\mu^{2\epsilon}$; 自能反项 $i(\delta_Z p^2 - \delta_m m^2)$;
顶点反项 $-i\delta_\lambda \lambda \mu^{2\epsilon}$.

$$[a_{\mathbf{p}}^\lambda, a_{\mathbf{q}}^{\sigma\dagger}] = -(2\pi)^3 2E_p \delta^{(3)}(\mathbf{p} - \mathbf{q}) g^{\lambda\sigma}.$$

$$\sum_{\lambda=0}^3 \epsilon_\lambda^\mu(p) \epsilon_\lambda^{\nu*}(p) = -g^{\mu\nu}.$$

$$D_{\mu\nu}(p) = \frac{-ig_{\mu\nu}}{p^2 + i\epsilon}.$$

$$\langle 0 | T A_\mu(x) A_\nu(y) | 0 \rangle = \int \frac{d^4p}{(2\pi)^4} D_{\mu\nu}(p) e^{-ip \cdot (x-y)}.$$

Quantum Electrodynamics

忽略规范固定项时的裸拉氏量:

$$\mathcal{L} = -\frac{1}{4}(F_{0,\mu\nu})^2 + \bar{\psi}_0(i \not{\partial} - m_0)\psi_0 - e_0 \bar{\psi}_0 \not{A}_0 \psi_0.$$

$$\psi_0 = \sqrt{Z_2}\psi, \quad A_0^\mu = \sqrt{Z_3}A^\mu,$$

$$m_0 = Z_m m, \quad e_0 = \frac{Z_1}{Z_2 \sqrt{Z_3}} e.$$

费曼规则:

- 入/出电子: $u(p), \bar{u}(p)$;
- 入/出正电子: $\bar{v}(p), v(p)$;
- 入/出光子: $\epsilon_\mu(p), \epsilon_\mu^*(p)$;
- 顶点: $-ie\mu^\epsilon \gamma^\mu$;
- 电子自能反项: $i(\delta_2 \not{p} - \delta_m m)$;
- 光子自能反项: $-i(g^{\mu\nu} p^2 - p^\mu p^\nu) \delta_3$;
- 顶点反项: $-ie\mu^\epsilon \gamma^\mu \delta_1$.

其中 $\delta_i = Z_i - 1$, $\delta_m = Z_2 Z_m - 1$ 。

Spinor helicity, massless

$$|i^\pm\rangle \equiv |k_i^\pm\rangle = u_\pm(k_i) = v_\mp(k_i),$$

$$\langle i^\pm | \equiv \langle k_i^\pm | = \bar{u}_\pm(k_i) = \bar{v}_\mp(k_i).$$

$$\langle ij \rangle = \bar{u}_-(k_i) u_+(k_j), \quad [ij] = \bar{u}_+(k_i) u_-(k_j).$$

$$\langle ij \rangle [ij] = 2k_i \cdot k_j = s_{ij}, \quad \langle ij \rangle = -\langle ji \rangle, \quad [ij] = -[ji].$$

$$\langle i^\pm | \gamma^\mu | i^\pm \rangle = 2k_i^\mu, \quad |i^\pm\rangle \langle i^\pm| = \frac{1}{2}(1 \pm \gamma^5) \not{k}_i.$$

$$[i|\gamma^\mu|j\rangle [k|\gamma_\mu|l] = 2[ik]\langle lj\rangle,$$

$$[i|\gamma^\mu|j\rangle = \langle j|\gamma^\mu|i], \quad \langle ij\rangle\langle kl\rangle = \langle ik\rangle\langle jl\rangle + \langle il\rangle\langle kj\rangle.$$

原卷以参考无质量动量 r 给出的光子极化矢量为

$$\epsilon_+^{\mu}(k, r) = -\frac{\langle r|\gamma^\mu|k\rangle}{\sqrt{2}\langle rk\rangle}, \quad \epsilon_-^{\mu}(k, r) = -\frac{[r|\gamma^\mu|k\rangle}{\sqrt{2}[rk]}.$$

Useful formulae

$$\frac{1}{A_1 \cdots A_n} = \int_0^1 dx_1 \cdots dx_n \frac{\delta(\sum_i x_i - 1)(n-1)!}{(x_1 A_1 + \cdots + x_n A_n)^n}.$$

$$\int \frac{d^d \ell}{(2\pi)^d} \frac{1}{(\ell^2 - \Delta)^n} = \frac{(-1)^n i}{(4\pi)^{d/2}} \frac{\Gamma(n-d/2)}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n-d/2}.$$

$$\begin{aligned} \int \frac{d^d \ell}{(2\pi)^d} \frac{\ell^2}{(\ell^2 - \Delta)^n} \\ = \frac{(-1)^{n-1} i}{(4\pi)^{d/2}} \frac{d}{2} \frac{\Gamma(n-d/2-1)}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n-d/2-1}. \end{aligned}$$

$$\begin{aligned} \int \frac{d^d \ell}{(2\pi)^d} \frac{\ell^\mu \ell^\nu}{(\ell^2 - \Delta)^n} \\ = \frac{(-1)^{n-1} i}{(4\pi)^{d/2}} \frac{g^{\mu\nu}}{2} \frac{\Gamma(n-d/2-1)}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n-d/2-1}. \end{aligned}$$

$$\begin{aligned} \int \frac{d^d \ell}{(2\pi)^d} \frac{(\ell^2)^2}{(\ell^2 - \Delta)^n} \\ = \frac{(-1)^n i}{(4\pi)^{d/2}} \frac{d(d+2)}{4} \frac{\Gamma(n-d/2-2)}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n-d/2-2}. \end{aligned}$$

$$\begin{aligned} \int \frac{d^d \ell}{(2\pi)^d} \frac{\ell^\mu \ell^\nu \ell^\rho \ell^\sigma}{(\ell^2 - \Delta)^n} \\ = \frac{(-1)^n i}{(4\pi)^{d/2}} \frac{\Gamma(n-d/2-2)}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n-d/2-2} \\ \times \frac{1}{4} (g^{\mu\nu} g^{\rho\sigma} + g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho}). \end{aligned}$$

$$\Gamma(\epsilon) = \frac{1}{\epsilon} - \gamma + \mathcal{O}(\epsilon).$$

两体无质量散射:

$$\frac{d\sigma}{dt} = \frac{|\mathcal{M}|^2}{16\pi s^2},$$

$$s = (p_1 + p_2)^2, \quad t = (p_1 - k_1)^2, \quad u = (p_1 - k_2)^2.$$

一般衰变率:

$$d\Gamma = \frac{1}{2m_A} \left(\prod_f \frac{d^3 p_f}{(2\pi)^3 2E_f} \right) |\mathcal{M}|^2 (2\pi)^4 \delta^{(4)} \left(p_A - \sum_f p_f \right).$$

Gamma 矩阵:

$$\text{tr}(\gamma^\mu \gamma^\nu) = 4g^{\mu\nu},$$

$$\text{tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = 4(g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho}),$$

$$\text{tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma \gamma^5) = -4i\epsilon^{\mu\nu\rho\sigma},$$

$$\gamma^\mu \gamma_\mu = 4, \quad \gamma^\mu \gamma^\nu \gamma_\mu = -2\gamma^\nu, \quad \gamma^\mu \gamma^\nu \gamma^\rho \gamma_\mu = 4g^{\nu\rho},$$

$$\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma \gamma_\mu = -2\gamma^\sigma \gamma^\rho \gamma^\nu.$$